



Personalized medicine: An alternative for cancer treatment

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ABSTRACT

The incidence of cancer continues to increase worldwide, resulting in significant physical, emotional, and financial challenges for individuals, families, communities, and healthcare systems. Cancer is projected to be responsible for approximately 10 million deaths in 2020, accounting for one in six deaths globally. Prostate, colon, lung, and breast cancers are the most common types of cancer. In India, it is estimated that there will be around 2.7 million cancer patients by 2020. Personalized medicine has the potential to offer an alternative approach to cancer treatment. Precision medicine, often known as personalized medicine, is a new cancer treatment technique that focuses on tailoring medication to each patient's specific genetic, biochemical, and lifestyle factors. The goal is to optimize tumor response while minimizing therapy side effects, resulting in improved patient care and quality of life. Personalized medicine allows for the creation of focused medicines that address specific gene mutations by leveraging knowledge about a patient's cancer, including its genetic makeup. Ongoing research seeks to detect gene modifications in diverse cancer types, produce novel diagnostic tools, and develop treatments that particularly target these genetic changes. In recent years, personalized medicine has achieved major advances in the treatment of solid tumors, with the promise to improve treatment precision, reduce side effects, as well as enhance outcomes for patients in cancer therapy. This review aims to objectively evaluate the transformation of cancer treatment, emphasizing the shift towards a more precise methodology.

Introduction

The concept of personalized medicine (PM) has emerged as a prominent research area to succeed in traditional treatment. It helps in determining the pattern of disease from the perspective of the patient's history and is helpful in the creation of medical devices and drugs unique to the patient. It is transforming the provision of healthcare to individual patients[1]. For several medical practitioners, this paradigm shift will manifest as advancements in foreseeing illness, followed by preventative measures such as medicines, testing for diagnosis, and treatment, including delivering targeted drugs for specific genes. Based on the patient's condition history, stage of disease, and susceptibility to

disease in a population, it is tedious to determine the diagnosis of a disease and precise treatment. The traditional or conventional methods developed by physicians may or may not cure a disease and can be quite tedious to implement for some diseases. Personalized medicine (PM) provides a significantly better treatment with its alternate or futuristic approach for the treatment of many diseases through understanding the patient's genetics and predicting the best treatment range. Next-gen sequencing is used for cost- and time-effective tumor DNA sequencing in research and treatment to target specific cancers along with patient genotypes [2]. It can classify individual patients into a unique or susceptible subpopulation for a particular disease and their response to treatment (drug medication such as synthesized medicines, herbal

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formulations) based on the stage of the disease, such as in different types of cancer. Fig. 1 shows the process of cancer development.

Cancer and personalized medicine

Cancer is a complex disease with a heterogeneous nature as it evolves within a patient or in a subpopulation, moderated by population, environment, socioeconomic factors, and genetic or cellular abnormalities. The abnormal and uncontrolled proliferation of cells without apoptosis forms tumors; these cells are referred to as neoplastic cells and are the cause of cancer. Even a single tumor can have a diversity of cancer cells; the heterogeneous nature of such cells can be observed through heterogeneous disease modulated by genetic, molecular, cellular, tissue, population, environmental, and socioeconomic factors, cell surface markers, growth rate, and other cancer progression traits during the progression of the disease. Such tumors can exist in the microenvironment of a complex system along with other cells of stromal, endothelial, or immune origin [3]. Other diseases can also worsen the risk of cancer if they are originated in same within the same organ with other complications such as contagious diseases (TB, COVID) along with cancer symptoms. Also the, complications can arise when medication for such diseases is given to the patient. Cancer can alter the extracellular matrix of tissues and organs at biochemical levels and alter the biomechanical functioning of the cell, as well as its architectural and topographical solidity. Any tissue or organ of the body can cause cancer from the somatic origin, having a similar invasion of cells at distant spread, or from hereditary origin, called genetic aberrations. Only a portion of the origin of cancer is understood, and most of its origin is due to causative factors like lifestyle, environment, or any infection-related origin [4].

Hereditary cancer syndromes of autosomal disorder due to faulty alleles are known, in which mutations inherited from offspring by parent genes are often involved in the triggering or onset of early cancer. Many genes involved in the suppression of tumors for protection against cancer are mutated, where DNA repair mechanisms fail to repair such oncogenes [5]. Commonly found syndromes with oncogenes include angiogenesis (blood vessel formation), Lynch syndrome, and breast and ovarian cancer syndrome. Cancer cells are not only produced due to some genetic inheritance, but they can also proliferate due to an affected lifestyle. The factors contributing to the development of various conditions include lifestyle decisions (e.g., smoking or intake of carcinogen substances, food habits, and exercise), environmental factors (such as exposure to some specific chemicals or radiation), and mutations arising during cellular division [6]. Fig. 2 summarizes the various types of

cancer along with their common symptoms.

The inclusion of PM in cancer research could lead to significant advancements in the fight against cancer. Prolonged patient therapy, complexity of stage, and interpatient variability (difference in response to cancer treatment) are the challenges faced by healthcare workers while monitoring patients. The identification of molecular alterations as targets for treatment has been a key factor in the progress of personalized medicine [7]. Genomic, proteomic, and metabolomic analyses have uncovered a variety of molecular alterations that can be exploited for personalized treatment. These include PIK3CA, ERBB2, FGFR, BRAF, and NTRK mutations or fusions. These discoveries have led to the development of targeted therapies that can specifically inhibit these molecular alterations, thereby halting the progression of cancer. Novel targeted agents are being developed for tumors originating from such identified mutations. Personalized treatment is specifically designed to target a specific alteration within the cancer. This is a significant advancement in the treatment of a group of patients, offering a personalized treatment that targets a specific alteration within the cancer. E.g., endocrine therapy in breast cancer, imatinib drug for chronic myeloid leukemia, and EGFR inhibitors in lung cancer have all demonstrated the potential of personalized medicine [8].

Symptomatic origin and development of cancer

An individual suffering from cancer possesses various symptoms. The signs and symptoms observed will frequently depend on the type of cancer the individual has. A lump (mass of tissue) in the breast, for instance, is a sign of breast cancer. An odd-looking mole is a sign of skin cancer. Typical signs and symptoms of many forms of cancer include excess weight loss, blood in the urine or stools, fatigue, and lumps that can be seen anywhere in the body. Cancer begins when normal cells undergo genetic mutations that cause them to grow uncontrollably and form a mass or lump called a tumor. These mutations can be caused by various factors, including exposure to carcinogens (such as tobacco and certain chemicals), genetic predisposition, and environmental influences [9]. Fig. 3 is the systematic representation of various causes behind the genetic changes in the individual.

In cancer, uncontrolled and abnormal division of cells very rapidly proliferate. Proliferate cells are the same as normal cells and the apoptosis process. Abnormal proliferative cells are known as neoplastic cells. There are two specific types of cancer one is benign tumors and another is malignant tumors. A benign tumor is a localized tumor because it is found in the upper surface of cells. It can easily be removed from body parts. It can't be transferred from one body part to another

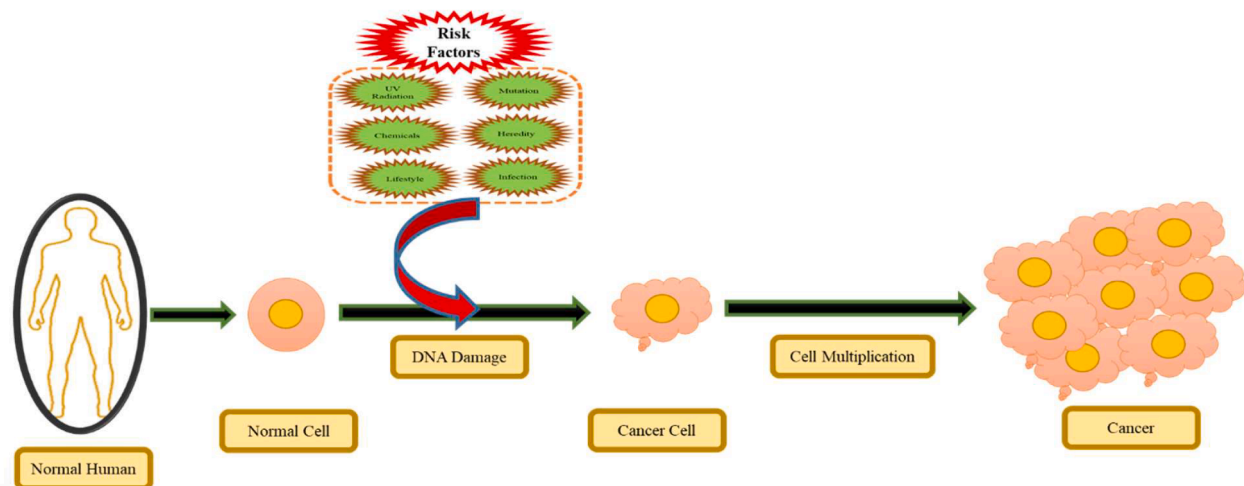


Fig. 1. A diagrammatic representation showing the development process of cancer.

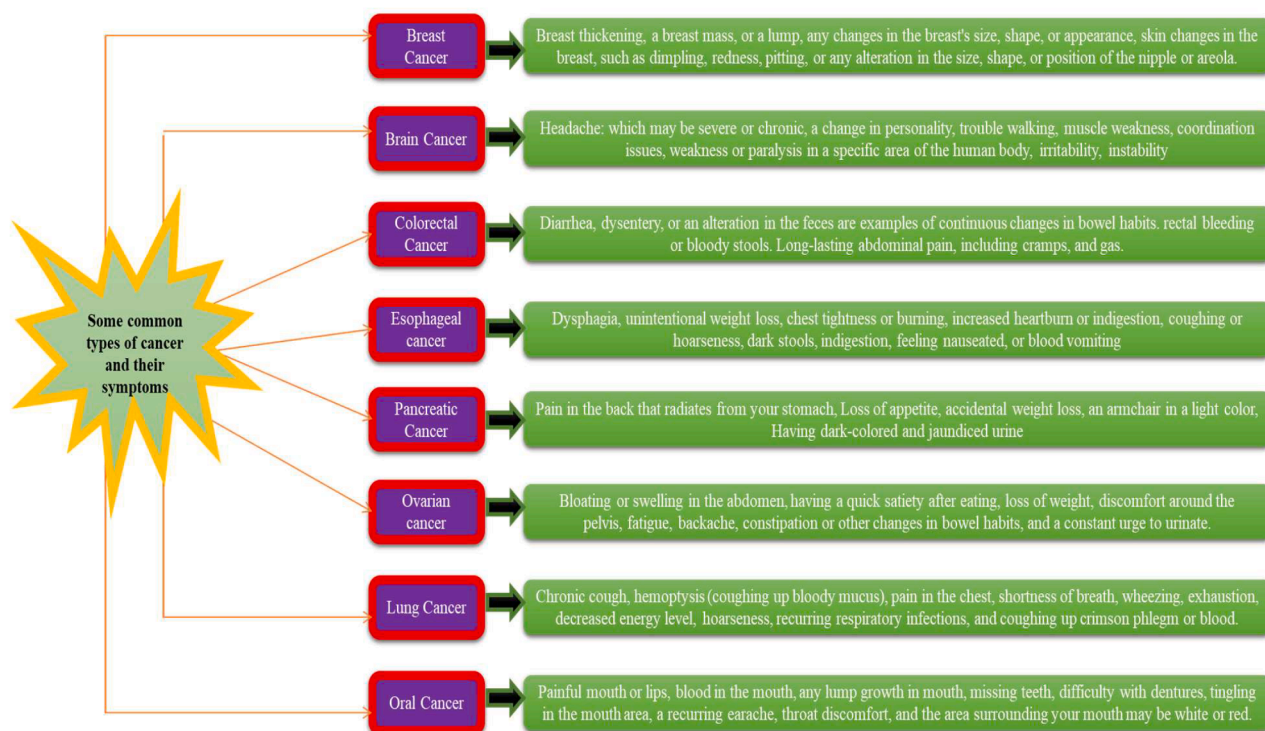


Fig. 2. A systematic presentation depicting the common types of cancer and their common symptoms.

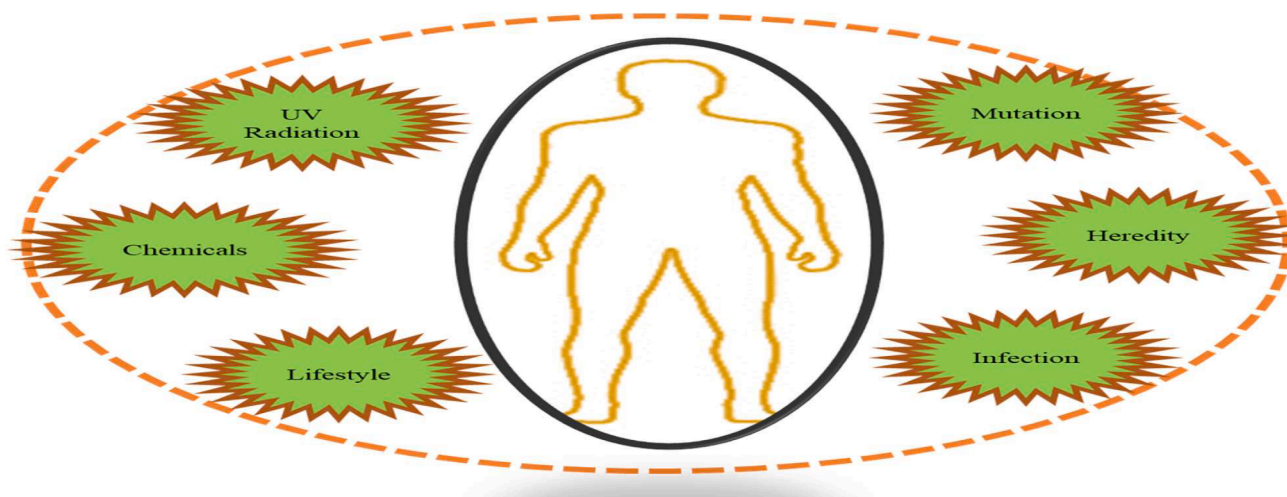


Fig. 3. Major causes behind the genetic changes.

body part [10]. Benign tumors have several types. First is Lipoma - an increase of fat cells in the body in the form of cluster form found in the skin and underlying muscle layer is known as Lipoma—Myeloma - increased muscle cells in the body, which is known as myeloma. Myeloma is a cancer of the plasma cells. Myeloma is known as multiple myeloma. Fibromass - increases fibrous cells of the body, it is mainly found in women's uterus. Malignant tumor - A malignant tumor is a dangerous cancer because it is a transferable cancer. It is transferred from one body part to another body part. It is not localized cancer. It has a metastasis process. It has several types. Carcinomas, Sarcomas, Leukaemia, Lymphomas. Carcinomas - Carcinomas are known as skin cancer. It is present in endodermal and ectodermal tissues on the surface of the skin. It is found on the upper surface of the skin [11]. It is a transferable cancer. It transfers from one cell to another cell. Sarcomas cancer - Sarcomas cancer is a Bone cancer, it is present in the bones of the

body. It is a transferable cancer because it can transfer from one body part to another body part. Lymphomas - Lymphatic system cancer is known as lymphomas. It is a transferable cancer because it can transfer from one body part to another body part. Leukaemia - an increase in abnormal WBC cells is called Leukaemia. It is a Blood cancer. It rapidly spreads all over body parts. It is a very dangerous cancer. The cancer-causing agent is known as a carcinogen. There are the following agents. Physical agent - Betal, Smoking, Drinking, Tobacco and Biological agent - 1- Parasite - Schistosoma parasite cause cancer 2- Viruses - Hepatitis B&C causes Liver cancer. Cancer-causing viruses are known as oncogenic viruses—radiation - X-ray, Gama ray, UV ray, Nuclear bomb [12].

Emergence of different types of cancer

The human body consists of trillions of cells. The cells are of such minuscule size that they are only observable via the use of a microscope. Cells aggregate to form the tissues and organs of our body. They exhibit a high degree of similarity. However, they differ in certain aspects due to the fact that body organs perform distinct functions. There exist around 200 distinct forms of cancer, which can be categorized based on their origin in the body, such as breast cancer or lung cancer [13].

Importance of classifying cancer diagnosis and treatment

Cancer detection is the most crucial stage as it drastically affects the survival of patients, but cancer is mostly detected at advanced stages due to unawareness or negligence by individuals [14]. Moreover, the affordability of treatments based on cancer type and stage for most individuals with economically weaker backgrounds may cause a step backward in the treatment of such diseases. Technological advances in the understanding of biological processes, such as powerful analytical molecular techniques, imaging, and histopathological methods with clinical trials, can provide substantial improvement in the evaluation of early-stage detection of cancer and advancement in cancer vaccine production (Fig. 4).

- 1. **Early Detection and Diagnosis:** Identifying the specific type of cancer and its origin is crucial for accurate diagnosis. Different cancers may require different diagnostic tests and screening methods. For instance, mammograms are used for breast cancer screening, while colonoscopies are used for detecting colorectal cancer [15].
- 2. **Tailored Treatment:** Each type of cancer behaves differently and responds to treatments in unique ways. Classifying cancers based on their origin (like breast, lung, or prostate) helps oncologists design

personalized treatment plans. For example, treatments effective for breast cancer might not be suitable for lung cancer [16].

- 3. **Research and Advancements:** Understanding the specific types of cancer is essential for ongoing research. Scientists study different cancers to develop targeted therapies and immunotherapies tailored to combat the unique characteristics of each cancer type [17].
- 4. **Prognosis:** The type and stage of cancer significantly impact a patient’s prognosis. Some cancers are more aggressive and spread rapidly, while others progress slowly. Classifying cancers provides valuable information about the likely course of the disease [18].
- 5. **Support and Awareness:** Cancer classifications help in raising awareness about specific types of cancer. This knowledge is essential for public education, support networks, and fundraising efforts aimed at finding cures and supporting patients [13,17].

Discription of different treatment methods for Cancer (Table 1)

Current global scenario

Globally, the worldwide burden of cancer prevalence as well as death is increasing at a rapid rate; this is due to a combination of causes, including population increase and aging, as well as shifts in the distribution and frequency of the primary cancer risk factors, many of which are linked to socioeconomic development.

In every nation on the planet, cancer is one of the main causes of death and a major obstacle to raising life expectancy. As per the World Health Organization’s (WHO) 2019 estimations, cancer ranks first or second in 112 out of 183 nations and ranks as third or fourth in 23 other countries as the primary cause of death under the age of 70. The fact that cancer is becoming more prevalent as the top cause of death is partially due to the notable declines in stroke as well as coronary heart disease mortality rates worldwide.

In India, one woman passes away from cervical cancer every eight minutes, and approximately one woman dies from breast cancer for

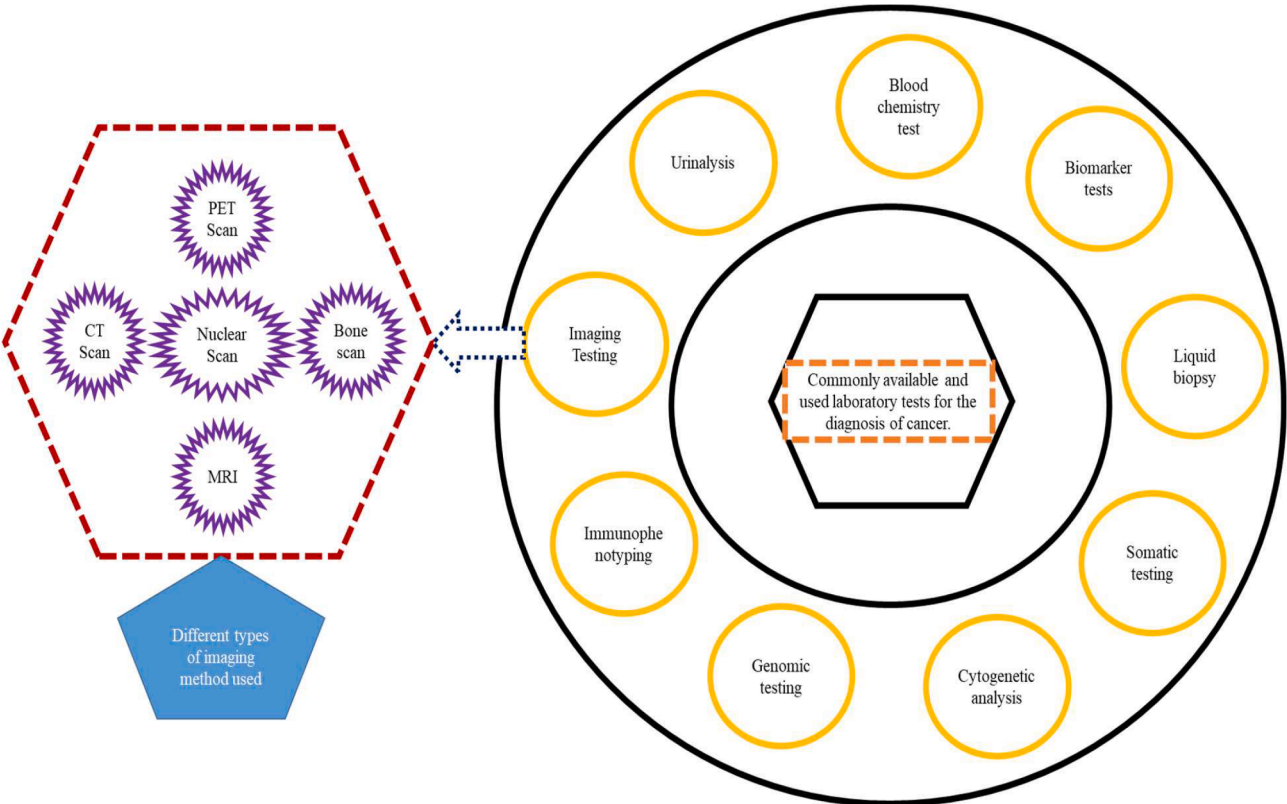


Fig. 4. Various types of tests which are commonly used for the diagnosis of cancer.

Table 1
shows the treatment type along with the medication.

Treatment Type	Description	Common Medications	References
Surgery	Removal of cancerous tissue, tumors, or organs to eliminate or reduce the cancer.	- N/A (Procedure-based)	[19,20]
Chemotherapy	Uses drugs to kill or slow down cancer cells' growth and division.	-Doxorubicin: Adriamycin- Cisplatin: Platinol- Paclitaxel:	[21]
Radiation Therapy	Uses high-energy rays to target and kill cancer cells or shrink tumors.	Taxol Fluorouracil: 5-FU - Radium-223: Xofigo- Iodine-131: Iotrex- Cobalt-60: Radioactive cobalt source	[22]
Immunotherapy	It boosts the body's immune system to recognize and destroy cancer cells.	- Pembrolizumab: Keytruda- Nivolumab: Opdivo- Ipilimumab: Yervoy- CAR T-cell Therapy: Kymriah, Yescarta	[23]
Targeted Therapy	Targets specific molecules involved in cancer growth, preventing the cancer from growing and spreading.	- Trastuzumab: Herceptin Imatinib: Gleevec- Rituximab: Rituxan- Erlotinib: Tarceva	[24]
Hormone Therapy	Interferes with the body's hormone production to slow or stop the growth of hormone-sensitive cancers.	Tamoxifen: Nolvadex- Letrozole: Femara- Anastrozole: Arimidex- Leuprolide: Lupron	[25]
Stem Cell Transplant	It replaces damaged bone marrow with healthy stem cells and is commonly used after high-dose chemotherapy or radiation.	Filgrastim: Neupogen- Plerixafor: Mozobil- Tacrolimus: Prograf- Cyclophosphamide: Cytoxan	[26,27]
Precision Medicine	Tailor's treatment is based on the genetic makeup of the cancer, identifying specific genetic changes for targeted therapies.	Larotrectinib: Vitrakvi- Entrectinib: Rozlytrek- Olaparib: Lynparza- Trifluridine/Tipiracil: Lonsurf	[8]
Palliative Care	It focuses on symptom management, providing relief, and improving the quality of life for patients with serious illnesses.	Morphine: MS Contin- Dexamethasone: Decadron Ondansetron: Zofran Haloperidol: Haldol	[28]

every two newly diagnosed cases.

It is estimated that tobacco usage causes up to 3500 deaths per day in India. In 2018, tobacco usage, including smoking, was the cause of approximately 3, 17,928 fatalities (approximately) among men and women [29,30].

PM for cancer treatment of various alignments

In India, various types of drugs and medicine (like herbal, Ayurvedic, or allopathic) have been given to suffering patients from ancient times in individualized forms [31]. These alignments are part of day-to-day cultural practices in the form of food items of plant and animal-based origins with different blends to the immunized body [3]. Personalized medications were based on herbs that are effective for certain symptomatic diseases and also immunized the immune system as a nontoxic adjuvant Fig. 5 shows the structure of the PM workflow.

Most of the diseases like cancer that mostly originated from Western countries were mere less or none in the Indian subcontinent. Due to personalized care from infancy to adulthood, many antimicrobial and anti-cancerous Ayurvedic, herbal, oil-based ointments were part of individuals' lifestyles [32]. The ointments mostly provide skin care for many epidermal diseases, and ingestion of herbal-based formulations provides immunity. They were also effective against dysentery, diarrhea or food poisoning, and many other health-related issues. Many herbal, Ayurvedic, or allopathic plant-based medications are a combination of chemical entities that are used in cancer treatments. Many of the medications like CHOP (chemotherapy) for non-Hodgkin lymphoma use cyclophosphamide, hydroxyl doxorubicin, Oncovin, CMF, i.e., cyclophosphamide, methotrexate, and fluorouracil [33]. The herbal formulations contain different active natural compounds that can be used in therapeutic treatments with lower adverse effects and are also synergistically used in treatments. Moreover, many of the cancer treatments become multidrug-resistant to cancer treatments to cancer cells, causing cross-resistance to chemotherapeutic agents. Nanoparticles can also be used to minimize toxicity to biological systems by targeting the use of these particles for treatment. These can be synthesized for different purposes, like antifungal and antibacterial agents. Also, nanoparticles can improve the diagnosis for improvement and management of different types of cancers [34].

Progress in drugs used in personalized medicine

The use of Pharmacogenomics for the betterment of personalized medicine by implementing genotyping technology to select and design drugs for individual patients or patients with similar diagnoses. The inter-individual genomic factor variability is a challenging aspect for predicting drug delivery and dosage, individual aspects of drugs such as improved efficacy, its activation, prevention measures, adverse reactions, further development or discovering more compatible drugs, and targeting the drug for cancer therapy models all needed to be considered in Pharmacogenomics. Pharmacogenes are determined by regulatory and coding regions within genes, which further influence pharmacology drugs for the biological systems of human beings [21]. The genome region decides the fate of the effectivity of drugs via modification of drug targets, interruption of sensitive drug biological pathways, alterations in their distribution or metabolism, etc.

The chemotherapeutic treatment of the drug is effective when the active compound reaches an effective concentration during treatment in tumor cells. Factors like the adsorption of the drug, its distribution and binding to proteins, metabolic activity status, effective transport within tumor cells, and readily removal from the body are crucial for effective treatment. Rapamycin and its derived forms are the drugs administered to treat renal cell carcinomas and soft tissue sarcomas, along with rare tumors. Several enzymes (CYP3A4, CYP3A5, and CYP2C8) and transporters (ABCB1) are involved in the metabolism and transport of rapalogues and that have known genetic variants that affect their activity [35,36]. Some tumors express high levels of these enzymes and transporters, which could reduce the intratumoral concentration and effectiveness of rapalogues. The tumors with mutations in the mTOR pathway, such as AKT, PTEN, TSC1, and TSC2, are more sensitive to rapalogue treatment. Hence, a three-layered approach for PM in the use of rapalogues for cancer treatment is being followed. Genetic testing of the patient to predict the tolerance and optimal dosing of rapalogues, pharmacokinetic monitoring of the blood concentration of rapalogues and review of co-administered drugs that might interfere with their metabolism, gene expression analysis of the tumor to assess the expression of resistance mechanisms and the activation of the mTOR pathway [37].

Dysregulation of the FGF (fibroblast growth factor) signaling axis is linked to oncogenesis, tumor development, as well as resistance to cancer treatment in a number of solid tumors [38]. FGFRs consist of FGFR5, which is devoid of an intracellular kinase domain, and the four

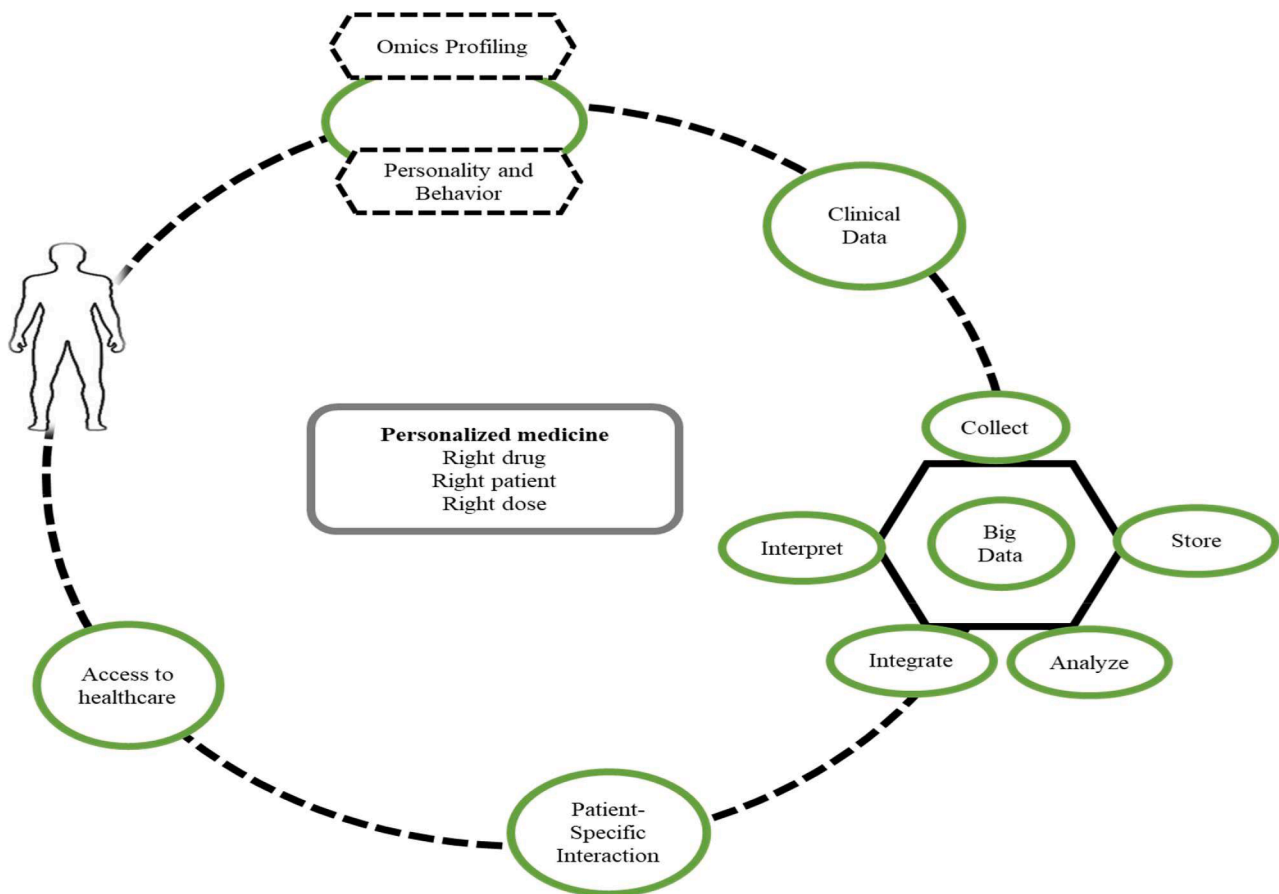


Fig. 5. A fundamental structure for personalized healthcare.

transmembrane tyrosine kinase receptors (FGFR1–4) that are highly conserved. FGF (fibroblast growth factor) activates each of these receptors, which helps promote cell division and survival. The abnormal FGFR signaling pathway has been suggested as a possible target for therapy in several trials examining various solid tumors, and numerous inhibitors are now being developed [39]. Fufibatinib is a new personalized cancer treatment that can significantly improve the prognosis for some patients with bile duct cancer. This drug targets a specific genetic alteration known as FGFR2 fusion, which is found in some cases of bile duct cancers [40]. This drug binds selectively and irreversibly to the ATP binding pocket of (fibroblast growth factor receptor) FGFR 1–4, resulting in the inhibition of FGFR-mediated signal transduction pathways, reduced tumor cell proliferation, and increased tumor cell death in tumors with FGFR 1–4 genetic aberrations. In a number of FGFR-driven human tumor xenograft models, oral administration of Fufibatinib resulted in a significant dose-dependent reduction in tumor size [41]. Furthermore, the tumor reduction was linked to a sustained FGFR inhibition that was dose-dependent. In comparison to chemotherapy, the medication likewise had mild adverse effects [42].

Use of AI in personalized medicine for cancer

AI is also being used in personalized treatment and management models from screening to diagnosis, early detection, and treatment to decrease mortality and recurrence in cancer cases. Deep learning-based AI automatic classification systems have been developed to differentiate between carcinoma and non-lactating mastitis in breast cancer. CNN (Convolutional Neural Network) algorithms have shown higher accuracy in the diagnosis and detection of breast cancer [43]. Continuous contrast-enhanced ultrasound allows radiologists to access the dynamic vascularization of vessels and tissues in real time, making it useful for

differentiating benign and malignant tissues using machine-learning models [44].

Various machine learning models have been used to predict lymph node metastasis, with the extreme gradient boost model being best for sentinel lymph node detection and preoperative clinical guidance [45]. Radiotherapy is often used alongside chemotherapy, but it has its limitations. 3D printed bolus is used in postmastectomy patients to increase dose effectivity. Seroma analysis after intraoperative radiotherapy is also being studied, but its effectiveness remains unclear due to variations in body reactions. Radiation-induced dermatitis can be predicted with high accuracy using the gradient-boosted decision trees model and random forest model as screening methods [44].

Genetic basis cancer biomarkers for treatment with personalized medicine

The advent of PM in cancer treatment has been driven by advancements in molecular testing and targeted therapies. These developments have led to improved outcomes for cancer patients with specific biomarkers. The initial diagnosis of cancer is based on the clinical history of the patient, cancer type and site in the body, and other laboratory investigations [46]. The patient's genome is analyzed for mutation along with a database search that predicts the severity of the mutation. The therapy is selected based on available former information from prior studies, consent of the patient for therapy preference, and availability of targeted agents or biomarkers that indicate mutation. Clinical trials of cancer patients with such novel treatments are beneficial in cases of poor prognosis via standard treatments. Also, the studies of such patients improve and benefit the subsequent cases in cost-efficient, improved outcomes and sharing of information [7,47].

Personalized medicine and related issues

Despite the significant progress, there are still limitations and challenges to overcome in the field of precision medicine. Tumor heterogeneity, resistance mechanisms, and complex molecular profiles can limit the efficacy of targeted therapies. However, the development of preclinical models, liquid biopsy techniques, and molecular tumor boards are promising strategies to overcome these challenges and optimize patient selection for personalized treatment. An evolving strategy for tumor prevention and therapy that considers the genetic variations inside and between tumors, the immunological milieu surrounding the tumor, the patient's lifestyle, and any associated morbidities is known as personalized or precise medicine in oncology [1].

PM has the capacity to modify the tumor's immune environment and target treatment toward the oncogenic causes of the tumor. The detection of this variation has been made possible by the availability of contemporary biomedical technology, such as proteomics, DNA sequencing, and wireless monitoring devices [48]. This has effectively exposed the necessity for some degree of medicine customization. Moreover, PM seeks to maximize tumor response while accounting for the toxicities brought on by therapy for each individual patient. In this method, the maintenance of the functioning of organs and, consequently, quality of life is paired with optimized tumor response. Furthermore, the ultimate goal of PM is to provide better patient care [49].

The difficulties that come with this reality in the future will be not only attempting to render individual characterizations more efficient but also creating and testing personalized medications so that they can demonstrate their efficacy. That's not to argue that treatments that are widely effective (i.e., the classic single-agent "blockbuster" medications) should be disregarded if found; rather, it means that they may prove to be extremely difficult to find in the future. There are some additional issues with personalized medicine that might be challenging to resolve soon. Large-scale data gathering, for instance, may be required to find characteristics that discriminate against groups of people who would benefit more from a certain solution. This could raise issues about privacy as well as the possibility that the information about those people will be misused for bad [49,50].

Conclusion

Many advancements in the treatment of solid tumor patients have been made possible in recent years by personalized care. In order to achieve a meaningful and substantial improvement in all clinical results, a deeper comprehension of molecular science is required. A strategy incorporating proteomics, metabolomics, and changes in DNA and RNA will be required to improve. The use of translational research utilizing organoids, xenografts, and liquid biopsy to assess molecular alterations brought on by clonal pressure resulting from target agent use or tumor heterogeneity would aid in the identification of mechanisms underlying resistance and raise the prospect of new combinations. It is imperative that further research be done to increase precision.

It is evident that the inclusion of a PPM perspective in cancer research and therapy could lead to significant advancements in the fight against cancer, given the disease's complexity and interpatient variability. This has already begun to be acknowledged in the current state of science and medicine through PPM research, PPM products, and FDA support; however, before PPM can be fully integrated into standardized care, a number of larger societal obstacles need to be addressed and overcome. The progress of personalized medicine in cancer treatment is a testament to the power of scientific innovation and collaboration. While challenges remain, the future of PM in oncology is bright, with the potential to transform cancer treatment and improve patient outcomes. As we continue to unravel the complexities of cancer at the molecular level, we move closer to a future where every cancer patient has a treatment plan tailored to their unique needs.

CRediT authorship contribution statement

Devendra Singh: Conceptualization, Investigation, Writing – original draft, Writing – review & editing, Supervision. **Vinay Kumar Dhiman:** Conceptualization, Investigation, Writing – original draft, Writing – review & editing. **Minakshi Pandey:** Writing – original draft, Writing – review & editing. **Vivek Kumar Dhiman:** Conceptualization, Investigation, Writing – original draft, Writing – review & editing. **Avinash Sharma:** Writing – original draft, Writing – review & editing. **Himanshu Pandey:** Writing – original draft, Writing – review & editing. **Sunil Kumar Verma:** Conceptualization, Investigation, Writing – original draft, Writing – review & editing. **Rajeev Pandey:** Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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